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# Trustworthy AI: AI developers' lens to implementation challenges and opportunities

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ABSTRACT

As organizations continue to embrace the use of artificial intelligence (AI) systems, it is crucial to ensure that these AI systems can be trusted. However, there is still a significant gap between research on trustworthy AI and its implementation in real-world applications. To address this issue, we sought to explore the perspectives of AI developers and the challenges they face in creating trustworthy AI systems. This exploratory study involved conducting interviews with 19 AI developers. We identified key challenges faced by AI developers due to the immature state of trustworthy AI, inconsistent global regulatory landscape, a lack of standardized definitions of key concepts, limited tools and standards for practical implementation in organizations. This paper provides recommendations for organizations to invest in trustworthy AI processes and practices, this includes building a foundation for trustworthy AI specific to their organization, adopting an organizational approach to trustworthy AI culture, and providing proper data infrastructures to support AI developers in creating trustworthy AI systems. By investing in trustworthy AI practices, organizations can prepare for evolving regulations and ensure that their AI systems are reliable and trustworthy.

## 1. Introduction

Organizations' use of AI technologies is on the rise. Today we see organizations use AI in their products, research and development practices, and also to enhance internal operations or even to better understand customers. The growing trend in using AI in organizations leads to the rising need for AI systems to be trustworthy (Choung et al., 2022; Marino et al., 2020). As described in the research literature, attention to trustworthy AI is important to promote responsible development and deployment of AI systems (Choung et al., 2022; Marino et al., 2020), with an emphasis on ensuring societal factors are considered throughout the AI system lifecycle. Raab (2020) and Morley et al. (2020) further highlight the contextual variations in trustworthy AI application across diverse organizational settings.

Experts have identified research gaps in trustworthy AI, including limited common definitions (Jobin et al., 2019; McCarthy, 2007; Morley et al., 2021; Vakkuri & Kemell, 2019), unclear standardization of trustworthy AI concepts and methodologies (Jobin et al., 2019), lack of consistent implementation of trustworthy AI practices (Jobin et al., 2019), and an overwhelming number of principles that lack practical clarity (Floridi & Cowls, 2022; Ibanez & Olmeda, 2022). Additionally, principles are often vague and unproven to support trustworthy AI in practice (Mittelstadt, 2019; Morley et al., 2020). In turn, there is a need to link AI principles, business processes, and data governance to specific procedures for both organizations and AI developers (Abrams et al., 2019; Ibanez & Olmeda, 2022; McNamara et al., 2018; Morley et al., 2021; Vakkuri & Kemell, 2019). The fluctuating trustworthy AI landscape limits an organization's ability to adhere to regulation while leveraging new AI opportunities (Floridi et al., 2018). However, while these research gaps underscore broad challenges in the current landscape of trustworthy AI, the specific research gap motivating this study is the disconnect between trustworthy AI concepts and its practical implementation. The pathway of turning trustworthy AI concepts into practical implementation is unclear. This is the core research gap we aim to address through further investigation with AI developers.

As we delve deeper into the landscape of AI, it is crucial to acknowledge the research gap that persist on the practical limitations of trustworthy AI implementation practices. Recent studies have emphasized the importance of having AI developers perspectives in bridging the gap between trustworthy AI research and its practical implementation (Abrams et al., 2019; Ibanez & Olmeda, 2022; McNamara et al., 2018; Morley et al., 2021, 2020; Orr & Davis, 2020; Vakkuri & Kemell, 2019). AI developers' play a key role in the entire lifecycle of AI systems, from creation and development to deployment. There is value

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in taking a AI developer centered approach to trustworthy AI research (Ibanez & Olmeda, 2022; Vakkuri & Kemell, 2019) and is necessary to gain a greater understanding of their perspectives (Ibanez & Olmeda, 2022; Orr & Davis, 2020). By doing so, we may also address the gaps in the current landscape of trustworthy AI and create a more secure and reliable future for AI systems.

AI developers have a unique set of roles and responsibilities in trustworthy AI. They have responsibilities for building, developing and deploying AI systems into production (Arrieta et al., 2020; Ibanez & Olmeda, 2022). In some cases, they can also be responsible for maintaining the AI system (Eitel-Porter, 2021). Recent literature highlights the need for AI developers to be a part of the trustworthy AI research and to help understand these under-explored research gaps (Ibanez & Olmeda, 2022; Vakkuri & Kemell, 2019).

Although organizational-level policies can contribute to establishing consistency in AI system design and development, a substantial portion of the decisions regarding the design and development of AI systems are made by AI developers (Ibanez & Olmeda, 2022). AI developers are individuals in an organization that work with AI systems to identify the trustworthy dimensions to build in the development of the AI systems (Ibanez & Olmeda, 2022). The risks of AI depend on complexities of an AI system and the domain it is being used in Floridi et al. (2018). Risks can occur if AI developers only consider trustworthy AI at the project ideation and development stages of the AI system lifecycle and/or overlook other trustworthy AI requirements in the AI system lifecycle (Graziani et al., 2022; Miller et al., 2017). Therefore, it is important for AI developers to have a clear understanding and consistent definition of trustworthy AI concepts and the necessary tools to embed them into the AI systems. Without this knowledge and preparation, there will be gaps in adhering to compliance and following organizational practices.

The notable research gap persists, an unclear pathway for translating these theoretical concepts into actionable steps for AI developers. This disconnect hinders the development of AI systems that align with trustworthy AI principles. Our research aims to bridge this gap by investigating AI developers' perspectives and the challenges they face in implementing trustworthy AI. This paper aims to provide a greater understanding to the following research questions:

- How do AI developers define and understand trustworthy AI concepts in the context of their work?
- What are factors that AI developers feel are barriers to adopting and implementing trustworthy AI into their work?

In order to tackle this gap, we conducted 19 semi-structured interviews with AI developers. We used the grounded theory method for coding our interview data. We categorized the emerging themes and reported the results in this paper.

This paper provides contributions to the discourse on the challenges encountered by AI developers during implementation of trustworthy AI systems, spanning from definition to actual deployment. Our research findings provide a comprehensive understanding of these challenges, providing insights to enhancing trust in AI systems. This paper advances the field by offering valuable recommendations, emphasizing the need for further standardization in defining trustworthy AI concepts for improved organizational practices to facilitate the adoption of trustworthy AI.

The manuscript is a structured approach, organized into distinct sections comprised of an examination of recent trustworthy AI research and a qualitative analysis involving interactions with AI developers. Subsequently, the paper delineates the research methods employed in our investigation. Our results are presented, followed by discussion of the implications and significance of our findings. The paper culminates a conclusion, summarizing key takeaways and potential areas for future research.

## 2. Related studies

Several studies have made an attempt to use a stakeholder approach to understanding trustworthy AI research. For instance, Ibanez and Olmeda (2022) conducted an exploration of the research gap existing between ethical AI principles and implementation at an organizational-level, with a focus on Spanish companies. Their study comprising of 22 interviews and 2 focus groups with organizations, brought to light the need for organizations to develop and use a specific code of ethics, provide a data-driven culture, considerations across an algorithm's lifecycle and training on ethical issues. This research provides a structure for our research design, to focus on the challenges faced by AI developers within organizations, rather than focusing on the organizations themselves. Minkkinen et al. (2022)'s study on responsible AI strategies categorizes the expectations derived from 15 interviews and document analyses. They found 5 distinct categories of responsible AI from work expectations, encompassing reproduction, translation, tension, scrutiny, and root development (Minkkinen et al., 2022). Vakkuri and Kemell (2019)'s developed a research framework for AI ethics, comprising of semi-structured interviews with 8 stakeholders. Their research contributes to an initial understanding of implementation, but a research gap remains in the literature concerning the perspectives and challenges faced directly from AI developers.

Trustworthy AI challenges impede the ability for effective implementation in organizations (Morley et al., 2020). Some existing tools and methodologies for implementing trustworthy AI, as investigated by Morley et al. (2021) and Floridi and Cowls (2022) demonstrate operational limitations and inherent variability for successful implementation. Morley et al. (2021)'s study concluded that the current tools and methodologies for trustworthy AI are not operationally effective.

While the technical components of trustworthy AI have received attention in the research, the perspectives of the AI developers who build these AI systems remain under-explored, Our research seeks to provide a greater understanding with insights into the challenges AI developers face when interpreting trustworthy AI concepts for trustworthy AI integration.

## 3. Methods

The primary objective of this research is to gain an understanding of AI developers' perspectives on trustworthy AI and its ability to implement in practice. Trustworthiness and ethics denote the reliability and integrity of the AI outcomes. Due to the focus on ensuring integrity, accountability and adherence to ethical standards, trustworthiness and ethics are concepts frequently used interchangeably in various scholarly discourses (Choung et al., 2022). This paper uses the concept of trustworthy AI to encompass both ethics and trustworthiness in AI systems. We employ a qualitative approach conducting semi-structured interviews for data analysis to bridge the gap between theoretical frameworks and practical applications.

In order to achieve our objective, we divided our research into distinct phases, as illustrated in Fig. 1. Our study followed a structured methodology. The initial phase of our research involved defining the scope of our investigation and gathering interview data. This phase, guided by the approach proposed by Lai and To (2015), laid the foundation of our in-depth analysis of AI developers' insights into trustworthy AI implementation.

### 3.1. Research participants and sampling procedure

We wanted to understand the views AI developers have towards trustworthy AI in practice, highlight the current challenges to trustworthy AI, and show areas of opportunity for future work. We developed a sampling plan to ensure a diverse and representative sample of AI developers for this research were invited, considering regional and

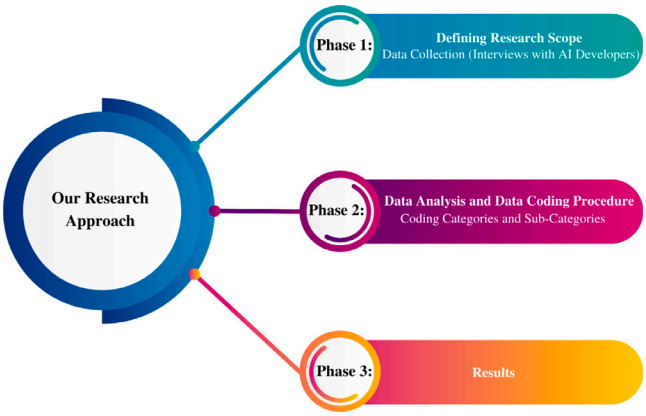


**Fig. 1.** Phases of data analysis process.

cultural differences, technology expertise, legal and regulatory considerations across various regions. Our sampling plan was used to ensure our research participants also had the specific AI developers traits that are relevant to this study's focus.

To better understand the AI developers' challenges in implementing trustworthy AI, we developed a visual representation of the AI system lifecycle, as seen in Fig. 2. Fig. 2 helps us construct the research questions to address all stages of the AI system lifecycle to ensure we gain an understanding of the practical limitations and challenges faced by AI developers. We modeled Figure 2 based on De Silva and Alahakoon (2022)'s Centre for Data Analytics and Cognition lifecycle work. Fig. 2 is a cyclical pattern beginning with the problem statement, followed by data collection and analysis, development, testing, deployment, finally monitoring. Fig. 2 is our breakdown of AI system lifecycle stages used to aid in crafting our research questions. The following are samples of the interview questions:

- What are the factors you feel are barriers and challenges to implementing trustworthy AI?
- As a AI developer, what do you feel is your expectation is to support trustworthy AI? Is this something that is organizationally-driven, individually-driven or both, and why?
- What is your interpretation on Transparency? How would you define Transparency as a AI developer? What are some methods/best practices to ensure Transparency in an AI system lifecycle?

Once we determined the interview questions, we selected the participants using criteria that combined several considerations. Technology expertise based on years of experience in the field. For example, novice (0–3 years' experience in the field), intermediate (over 3–5 years' experience in the field), and expert (over 5 or more years' experience in the field). We also considered the legal and regulatory environment in each region. For example, England using the AI Act and General Data Protection Regulation (GDPR), Canada has the draft AI and Data Act, United States has various state-driven laws and regulations, where India does not have their own laws on AI. Examples of AI developers include: data scientists, applied scientists, engineers, AI leads, and others. In this paper, we refer to all participants as AI developers.

We had each interview participant complete consent forms to ensure informed consent, while ensuring research data collection and storage practices applied. Our research data consists of interview data from 19 AI developers across technology organizations. The research participants are diverse as they vary in gender and ranged from executive level to entry level positions across the organizations, as outlined in Table 1. The interviews were in digital format, recorded and transcribed. We coded and assessed the research data using NVivo® 12 software.

**Table 1**
Research participants.

| Domain | Organizational level | Location |
|---|---|---|
| Sales/Go-to-market | Executive | United States |
| Sales/Go-to-market | Director | India |
| Sales/Go-to-market | Manager/Lead | India |
| Sales/Go-to-market | Manager/Lead | India |
| Operations/Technology | Data Scientist/Developer | India |
| Technology | Data Scientist/Developer | India |
| Operations/Technology | Director | India |
| Human Resources | Director | United States |
| Operations/Technology | Data Scientist/Developer | Canada |
| Operations/Technology | Manager/Lead | United States |
| Technology | Director | India |
| Operations/Technology | Executive | United States |
| Sales/Go-to-market | Manager/Lead | Canada |
| Operations/Technology | Data Scientist/Developer | India |
| Operations/Technology | Data Scientist/Developer | United States |
| Operations/Technology | Data Scientist/Developer | United States |
| Technology | Data Scientist/Developer | India |
| Human Resources | Data Scientist/Developer | England |

After our interview data was collected, the interviews were transcribed and we gathered about 320 pages of interview data. We converted our research data into a text file and removed any personally identifiable information (PII) data of the organization and our research participants. PII data is information that can identify an individual of which the information is about (Narayanan & Shmatikov, 2010). We analyzed our data to identify patterns, correlations and insights to better understand AI developers' perspectives to trustworthy AI. Our sampling plan helped ensure that our research is a well-rounded taking into account the diverse factors mentioned while maintaining ethical standards.

To better understand our data, we then applied grounded theory approach to analyze the data. We analyzed our research data logically and systemically (Lai & To, 2015). The grounded theory approach collects data in the form of interviews, codes the data into categories and subcategories, provides refinements and comparison to the categories of coding based on the data. It drives thematic findings from the grounded theory developed. We also conducted keyword analysis on our research data. The following section below will dive into these data analysis processes.

### 3.2. Analytic approach

#### 3.2.1. Grounded theory method

In this research, we employed the grounded theory approach, a qualitative method widely acknowledged for its effectiveness in deriving meaningful insights from text data (Heath & Cowley, 2004; Strauss & Corbin, 1998). Grounded theory by Strauss and Corbin (1998), is a systematic research methodology that facilitates the development of theory grounded in empirical observations, with a particular emphasis on exploring concepts and themes based on qualitative data (Arnold et al., 2017; Heath & Cowley, 2004; Lai & To, 2015; Witz, 2007). This methodology facilitates a systematic exploration of concepts and themes that emerge from the experiences and perspectives shared by the AI developers. We began with open coding to breakdown the transcripts and identify initial concepts. These codes were then grouped into categories through axial coding to reveal the data patterns. Finally, selective coding was established as a primary core category to represent the core themes of AI developers. This iterative coding process in grounded theory enables us to develop theory that is grounded in insights from our research participants.

#### 3.2.2. Data analysis

Our study adopted the grounded theory method to analyze the empirical data collected. This approach enabled us to discern relationships and patterns within the data, aligning with the foundational tenets of

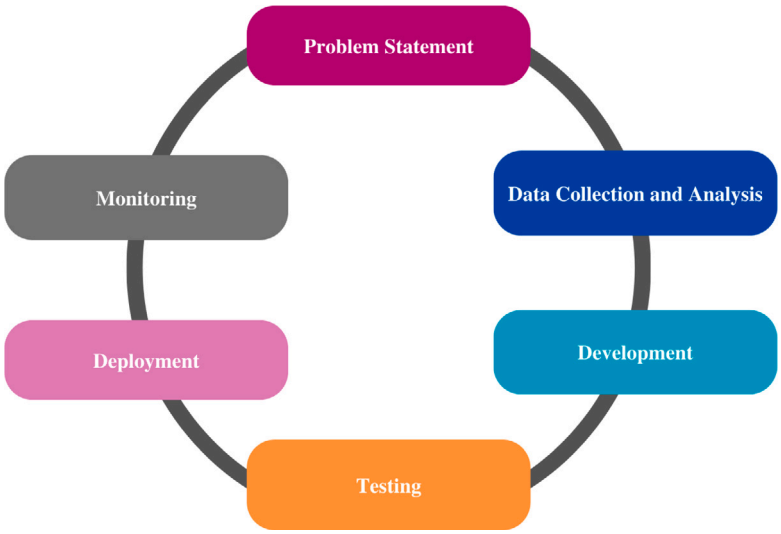


**Fig. 2.** AI system lifecycle.

**Table 2**
Coding categories.

| Coding categories | Coding sub-categories |
|---|---|
| AI system lifecycle | Types of AI systems |
| | Types of data |
| | AI system monitoring |
| | End users and customer involvement |
| Trustworthy AI Concepts and Techniques | Transparency |
| | Interpretability and Explainability |
| Trustworthy AI Landscape | Governance and Standardization |
| | Maturity of trustworthy AI |
| | Regulation |
| Importance of Data | Data interpretation |
| | Data reviews in development |
| | Data reviews in production |
| Purpose-Driven | Organizationally driven |
| | Individually driven |

grounded theory (Walker & Myrick, 2011). Our analysis consisted of multiple phases, with the second phase involving a coding procedure to identify key themes within the data set to facilitate the interpretation of our findings (Lai & To, 2015).

### 3.2.3. Coding procedure

The coding procedure, an essential component of grounded theory method, following a qualitative coding approach. As delineated in Table 2, the coding and sub-coding categories were derived from a detailed analysis of interview transcripts, allowing for a systematic extraction of key concepts and themes.

### 3.2.4. Analysis through tokenization, frequency and keyword extraction

To strengthen our results, we also conducted analysis through tokenization, frequency and keyword extraction. We removed all stop words from the data. Stop words are commonly used words, for example “as”, “an”, “the”, “um”, “so”, etc. We cleansed the data directly in Netlytic software. Netlytic is a cloud-based text and social analysis platform (Gruzd, 2022). Netlytic is a tool used for conversational data to identify and visualize themes throughout the conversations (Gruzd et al., 2016). Gruzd (2022) uses the dictionary-based approach to derive the topical analysis.

We then ran this cleaned data set through Netlytic’s keyword analyzer to pull the keywords and frequencies. The first step in this process was tokenization, which involves breaking down the text into its constituent tokens, typically words or phrases. This facilitates further analysis by converting the text into a structured format. Our text data was tokenized into individual words and phrases, making it more amenable for subsequent analysis, allowing for the effective handling of textual data. The keyword extractor was used as our method to identify popular topics in the data set. Netlytic also uses stemming technique on the research data. Stemming is a technique used to simplify words using the same root by removing derived suffixes and forms (Lovins, 1968). Netlytic counts how often each word is mentioned throughout our data set.

The keyword extraction analysis showed that participants emphasized several recurring themes. In this summary, Section 4 present the main findings from this thematic analysis, followed by a discussion on how these findings impact researchers and AI stakeholders.

## 4. Results

In order to understand the relationship between theoretical concepts of trustworthy AI and its practical implementation, we constructed a visual illustration presented in Fig. 3. This illustration delves into the interconnections of established trustworthy AI concepts with practical perspectives of AI developers. The insights gained from this representation are to better understand the implications of AI system development and deployment, highlighting the challenges and opportunities encountered by those directly involved in creating AI systems.

Fig. 3 highlights specific areas where AI developers’ experiences should be used further to shape the integration of trustworthy AI concepts. More specifically, the literature reveals that interpretability, XAI, and transparency share similar roles and responsibilities (Abedin, 2022). Our results found AI developers employ these terms interchangeably without a unified or consistent set of definitions. The diversity of

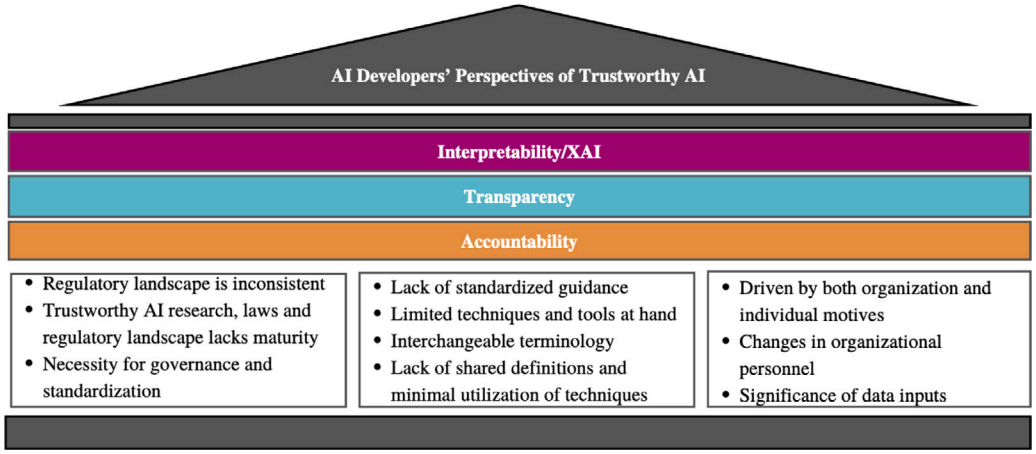


**Fig. 3.** High-level relationship mapping of our results.

definitions and uses among AI developers regarding these trustworthy AI concepts further complicates their practical implementation. This disparity highlights the need for clearer definitions and communication between researchers and practitioners in the field.

Fig. 3 is linked to the following Sections 4.1 to 4.6, providing insights into the real-world implementation of trustworthy AI systems. Accountability is a central responsibility AI developers emphasize the importance of accountability at both the organizational and individual levels. This connection warrants an in-depth exploration in subsequent sections of this paper.

Our research reveals that AI developers emphasize the necessity for governance and frameworks, an understanding of data inputs, challenges presented by an inconsistent regulatory landscape, and the lack of standardized guidance or well-defined practices.

The remaining sections in this paper will explain these high-level relationships of our results, providing a nuanced and comprehensive understanding of the interplay between the roles and responsibilities of trustworthy AI and the perspectives of AI developers in the landscape of AI systems.

### *4.1. Regulatory landscape of trustworthy AI*

An inconsistent global regulatory landscape can increase greater uncertainties regarding successful implementation of trustworthy AI. It was evident that AI developers acknowledged the maturity of trustworthy AI shapes uncertainty in the ability to implement trustworthy AI. The regulatory landscape of trustworthy AI is not unified globally:

> Quote #19: "It's not yet mature enough for us to say [that] we have a standard way of looking at things and this is the process".

> Quote #2: "Well, ethics in AI right now is still in major infancy".

> Quote #2: "Rest assured that we aren't using their private data or information for stuff that we aren't supposed to be especially in areas and countries where there is no clear laws around data privacy. I mean, Europe has GDPR. India doesn't have any".

The inconsistent global regulatory landscape was found to cause a lack of focus and direction for AI developers to create common awareness or consistent execution:

> Quote #3: "It is definitely a big challenge because the AI systems that have been built, I've seen or have been working with the people building the AI system there is not any focus on the regulations. The biggest challenge is people are not actually aware that they need to look at the regulation aspect of it".

> Quote #16: "What the rules are, because right now I'm putting my own ethics in place as an employee, I'm representing myself in what I would want".

The complexity of AI systems also influences its' interpretability (Adadi & Berrada, 2018). The research data also highlighted how building AI systems alone are complex in nature:

> Quote #2: "...but getting computers to think like humans. When you look at ethics and how we think as humans, it's not just ethics, right? There's so much involved. It's the relationships, the emotions and all of those factors play a huge part in how we make decisions, right. We are only trying to address certain portions of it with ethics and stuff. But there are many other fundamental layers involved in us. Taking decisions as human and training all of that into a system. I honestly don't know how".

> Quote #19: "It's not a one-size-fits-all and even like a single AI system that might have multiple users, how you use that AI system within your particular use case defines how you would track some of these other metrics like ethics or bias or you know quality. It is subjective to the problem you're trying to solve...".

According to a study by Jobin et al. (2019), 84 documents regarding trustworthy AI were analyzed and it was discovered that there were no correlations between the interpretation or usage of these principles and the domains they were applied to. Our research data supported that the challenges in trustworthy AI may cause teams or individuals having to:

> Quote #16: "...make it up as we go, and if you're a trustworthy company like this organization, we are making it up...".

Morley et al. (2020) found that AI practitioners tend to choose trustworthy AI tools that are either convenient for them or align with their personal understanding of trustworthy AI. However, their selection may not necessarily be the most ideal choice from a societal perspective (Krishnan, 2019).

### *4.2. Trustworthy AI should be organizationally and individually driven*

In order to understand the origins where trustworthy AI support should stem from, we asked AI developers. We investigated whether this support should come from organizations, individuals or a combination of both. Our findings revealed that 100% of AI developers (19 research participants) believe that trustworthy AI should be organizationally driven. Additionally, 84% AI developers (16 research participants) of respondents emphasize the significance of trustworthy AI being a joint effort between the organization and the individual:

> Quote #7: "It's both organization and individuals, decision makers and stakeholders think everybody is aware... It's a share of responsibility".

According to Mittelstadt (2019), there is a connection between the limitations of AI developers and the organization they work for. For trustworthy AI to be effective on a larger scale, it's crucial for any failures to be linked to the organization as a whole. Abedin (2022) emphasizes the importance of clear expectations and defined responsibilities for all stakeholders in decision-making. As AI developers are at the core of AI systems throughout their lifecycle, Ibanez and Olmeda (2022) and Vakkuri et al. (2019) stress the crucial role and responsibilities of AI developers in ensuring trustworthy AI implementation.

### 4.3. Defining trustworthy AI concepts

As Morley et al. (2020) identified, internal employees in an organization should first contextualize trustworthy AI principles, identify the appropriate tools and put them into the design of the AI system, and document how the process was followed. Inconsistency in how trustworthy AI concepts are defined can impact the ability to ensure trustworthy AI is adequately or appropriately implemented. Our research data found some AI developers to use trustworthy AI concepts interchangeably:

> Quote #7: "We have transparency, we have interpretability, and what was the first concept. These three concepts are very close to each other in my understanding, I think sometimes people are using these almost interchangeably".

This finding is aligned with the research literature, Abedin (2022) highlights researchers often use interpretability, XAI and transparency interchangeably, while other researchers have delineated these concepts from one another. It is concerning that there is no standardized definition for the concepts in trustworthy AI, as this leads to confusion and inconsistent usage. This issue has been noted by researchers such as Jobin et al. (2019), who emphasize the importance of clear and comprehensive definitions. Moving forward, it is crucial for further research to address these discrepancies and provide guidance for AI developers. Our own research has revealed inconsistencies in the interpretations of concepts such as interpretability, XAI, and transparency within the realm of trustworthy AI. Consistency in the concepts of trustworthy AI is crucial for AI developers when implementing trustworthy AI into AI systems as it provides a clear and well-defined framework. Consistency in trustworthy AI concepts can support clarity in AI developers' commonly understanding the goals and requirements to embed in their AI systems, avoid legal and regulatory issues, enhance user trust in AI systems, AI developers can build AI systems that work to uphold standards consistently.

#### 4.3.1. Interpretability and XAI

We focused on the trustworthy AI concepts of interpretability, XAI and transparency as these concepts were the most readily used in the research literature. For example, interpretability publications in the literature have more than doubled from 2012 to 2019 (Arrieta et al., 2020). We pulled the terms interpretability and XAI from the research data and ran a keyword extraction analysis to look for patterns.

Although interpretability and XAI are generally viewed positively by AI developers, our research revealed that AI developers still have a limited understanding and maturity when it comes to these concepts. This finding is consistent with our previous results on the immaturity of trustworthy AI concepts, including interpretability and XAI:

> Quote #19: "I think explainability is actually probably one of the areas that I feel personally like it hasn't matured. For example, if you're using an AI AI system to assess. What is it kind of like loan recipients, for example, I think generally you'll kind of get a binary answer without kind of getting that feedback loop".

#### 4.3.2. Transparency

Transparency is seen as a pre-requisite for algorithm(s) in AI systems to be trustworthy (Anagnostou et al., 2022). Similar to interpretability and XAI, there were various conceptions of transparency among AI developers:

> Quote #19: "Transparency would come in the form of exposing the data sets that we use to train the AI systems".

> Quote #7: "Transparency is mainly related to AI systems being kind of non-discriminatory".

> Quote #14: "Think it's a hard question for me to answer... [Do] we have security in place basically".

With various conceptions of transparency from our research data, AI developers generally are aware of the issues and try to implement degrees of transparency with limited tools in the market (Eitel-Porter, 2021; Ibanez & Olmeda, 2022; McNamara et al., 2018; Morley et al., 2020). There is also recent literature that identifies how the approach to trustworthy AI in practice requires judgment and depends on the organizational setting (Morley et al., 2020; Raab, 2020).

### 4.4. Types of AI systems

A discernible pattern in AI developers with successful implementation of trustworthy AI, relates to the significance of the AI system's nature and the methodologies applied. AI developers' decision-making process hinges on the characteristics of the AI system in accordance to the problem they aim to address. AI developers determine the suitable infrastructure for constructing AI systems, which should successfully undergo testing and validation phases. Additionally, during AI system development, AI developers engage in supplementary techniques. AI developers fine-tune the AI system, adjusting its hyperparameters:

> Quote #15: "the number which is give into me may not be sufficient for me to build up an AI system, and then I need to use some more of my own internal data science techniques to recreate it".

It is worth noting that AI systems vary in complexity (Adadi & Berrada, 2018; Biran & Cotton, 2017; McCarthy, 2007; Mueller, 2019). In this context, the AI developer assumes the role of a decision-maker when determining the requisite type of AI system:

> Quote #12: "If the simple AI system works well, if it doesn't, then you need to look at more complex AI systems now".

Machine learning (ML) employs computational techniques to enhance performance and make predictions (Heidecker et al., 2024). ML constitutes a subset of AI systems, and it can be categorized into supervised, semi-supervised, or unsupervised learning (Aziz & Dowling, 2019). Supervised ML learning involves input data to evaluate the output of an AI system, while unsupervised ML learning exclusively utilizes input data about data structures (Ayed, 2022; Aziz & Dowling, 2019). Evidently, the choice of ML learning type is linked with the methodologies applied in the development of AI systems:

> Quote #18: "When we talk about supervised and unsupervised in a different manner. Supervised we know we know what is a feature that we need to predict that comes a regression and classification couple of things. I've used a lot of regression and classifications like linear regression, random forest. These kind of things. And when we talk about the unsupervised things where we don't know what to do with the data and we need to do some data mining. In that case there are clusters clustering techniques".

### 4.5. Data in the AI system lifecycle

#### 4.5.1. Importance of the data

ML has changed the way patterns are extracted from the data (Felzmann et al., 2020), this relationship of data in an AI system was prominent throughout our research data. The topic of data is displayed across several research patterns which include: (1) the importance of your data and, (2) how data is a barrier in the AI system lifecycle.

> Quote #17: "…but our knowledge on that domain like knowledge on the field which you have working is a barrier in the initial steps for example, how we understand people data is completely different from other non living things or sales data or telecom data.[Having the] complete knowledge on the data which we are dealing. It could be a barrier in the initial steps…".

The AI developer's knowledge of the data going into the AI system was found to hinder the AI problem and development processes and/or impact the AI system's performance:

> Quote #10: "I just think the kind of evaluation of these AI systems are still relatively subjective like we can talk about like fairness, metrics, we can talk about. You know, some of those precision and false positive rates and things like that. If it can ultimately align towards like where security is today, I think it still has a bit of a ways to go. Because like I said, there's numbers. There's certifications that you can always align your product towards and it becomes also a marketing kind of vehicle…".

As AI systems continue to increase in complexity and configurations with large amounts of training data, these AI systems become impossible to trace each step (Felzmann et al., 2020). Felzmann et al. (2020) points out concerns should be considered for engineers to understand what data is used for what conclusions or decisions by the AI system. Traditionally algorithms in AI systems have rules programmed with weights attached to specific data points.

#### 4.5.2. Types of data

In the context of AI system development, AI developers require a deep understanding of the data they work with. This understanding serves as a foundational step in the process of integrating data into AI systems. 89% of AI developers emphasized the choice of AI systems for specific problems is contingent upon the characteristics of the data, such as its volume and complexity:

> Quote #18: "…the type of AI system s we are using for some particular problems again plays a crucial role. Depending on the suppose if there is a lot of data and the people doesn't know what to do there are a lot of unsupervised techniques like clustering many things but we need to get some fair idea which AI system we have to use and to solve that particular problem with the existing data. two things that are dependent. one is data and one is the building and choosing and building of the AI system".

This underscores the significance of selecting the appropriate AI system to help solve the problem at hand, talking into account the availability of data. Data, as a fundamental component in the AI system lifecycle, plays a pivotal role in shaping the AI system itself. The quality and suitability of data concerning its type, structure, and volume have a direct bearing on the AI system's performance. AI developers identified the intricate interplay between the data and the AI system, where the former is perceived as a dependency. Specifically, when AI developers are faced with large data sets, particularly in cloud environments, the complexities of the data and limitations of local infrastructure influence the AI system's development:

> Quote #7: "…going to cloud platforms has lot of dependency issues like permissions the data permissions. If the data is too big we can't handle it in the local infrastructure".

> Quote #18: "We can't download and we can't manually inspect those. Why we are getting some errors. there are some gaps where the Data Protection Act regulate restricts us to see what is happening inside the data. we need to just keep that everything in the protected place and we just need to do a through scripting itself. We just need to see that takes a bit of a difficulty without understanding properly that that that creates a gap".

There was notable recognition from AI developers on handling data that is restricted by privacy and regulatory concerns, indicating that proper data management can be challenging, which may result in gaps in the underlying data. Effective data preparation is integral to the AI development process and should ideally involve collaboration between AI developers, data domain experts, end users and customers of the AI system. AI developers are tasked with conducting in-depth parameter and feature analyses to ascertain the specific data requirements for the AI system:

> Quote #18: "We can get the best features out of the existing one. Suppose there are some 300 or 400 features that are coming in the data set. We can't pass everything directly to the AI system we just need to do some feature engineering and feature selection. we have a lot of techniques to do that feature engineering and feature selections".

This process of feature selection assists in optimizing the data input in the AI system, ultimately enhancing its performance and relevance of the AI system. The relationship between data and the development of trustworthy AI systems, underscores the role of data in AI system selection, development, and overall performance. The challenges associated with data management and regulatory constraints further emphasize the importance of comprehensive data strategies and collaboration with data domain experts in the AI system.

#### 4.5.3. Data interpretation

One of our findings revolves around the intricate relationship between data and the AI system. Although our research questions did not explicitly inquire about the concept of 'data', recurrent theme emerged, by the AI developers on the importance of high-quality data in the context of AI systems:

> Quote #15: "Understanding the data, see the customer is the person who has his data, who is passing the data to us so. There should be some see whenever we are starting the data, whenever we find some discrepancy in the data. See the only point of contact is a customer. When we have the more interaction with the customer we can get more insights of the data. The more control we have on the data, the better is the results or the better we can".

> Quote #3: "There are many challenges. One major thing that I see is there isn't enough data for building the AI system s. Some people argue here this is all the data we have, so let's do a work around and let's try and add rules to it make it a rule based so eventually it turns out from being a data science AI system to something that's a hybrid, which gives U.S. data signs and rules which kind of generates bias because the rules are defined by people who think are probably they know their business better, but they're understanding may not be the entire truth because there's definitely always hidden stuff in every case of".

AI developers need to possess a comprehensive understanding of the underlying data. This understanding, we found, often did not originate within the ranks of the AI developers themselves. Instead, it originates from the customer and/or end user. Our research found a limited grasp of the data underpinning the AI system was strongly correlated with a range of adverse outcomes, including sub-optimal system performance, protracted production timelines, and diminished value to the end users/customers.

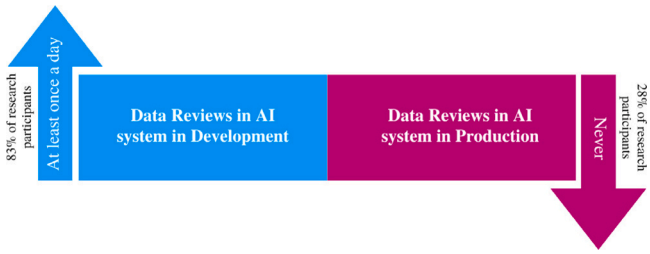


**Fig. 4.** Comparisons of data reviews in AI development versus production.

#### 4.5.4. Data reviews for AI systems in development versus production

In our pursuit to address the discerned data requisites of AI developers within the realm of AI, we analyzed the patterns of data reviews throughout stages of the AI system lifecycle. As showcased in Fig. 4, it became clear during the development phase of an AI system, a striking 83% of AI developers actively engage in the practice of daily data reviews. This highlights the role it plays in ensuring the AI system's integrity, functionality and performance. When we turned our attention to AI systems in production, our findings unveiled the practices surrounding data reviews within this phase ranged from sporadic, or even negligible. The variations in approaches used by AI developers speaks to the multifaceted nature of AI deployments, reflecting the unique requirements and challenges faced by AI developers. Notably, a substantial contingent of AI developers, comprising 28% of our cohort, acknowledged that data reviews once the AI system transitions into a production environment. This highlights another gap of inconsistency and an opportunity for further investigation of sustaining data oversight and quality assurance throughout an AI system lifecycle, specifically pre and post deployment. This finding reveals the importance of enhancing the development and deployment of AI systems but also a need to shape future strategies and best practices for data reviews in AI systems.

#### 4.5.5. AI system monitoring

AI developers identified the need for systematic and sustained monitoring of AI systems, which has been identified as an opportunity to enhance the reliability and robustness of AI system outputs. Ensuring the AI system has monitoring to provide a continuous and real-time assessment of the AI system's performance:

> Quote #6: "We need to observe it for any issues right? obviously in our AI system we have it is a machine learning operations kind of scenario what we have done is we have implemented lifecycle steps that are monitoring the input to you know test data to that AI system that we understand that how our AI system performance is whenever we see that our AI system performance is not up to then we go back to the data".

> Quote #12: "The monitoring has to be done on a daily basis but suppose if you keep it like 5%, there can be only 5% change in your evaluation metrics during the time frame. you need not to look every day. But suppose if at a on a certain day your evaluation metrics falls more than 5% of what you're training accuracy or training evaluation metrics was then you need to look into that, it is a case by case issue".

Through ongoing monitoring, AI developers gain knowledge in the behavior and functionality of AI systems. This, in turn, equips AI developers with the knowledge necessary to promptly detect and respond to any indications of performance degradation or drift in the AI system. The AI system monitoring was found to be a key component in the AI system development and deployment. By proactively assessing the AI system's performance, AI developers are better prepared to take timely corrective actions, thereby ensuring that the system consistently operates efficiency. Consequently, the implementation of AI system monitoring bolsters the performance of AI systems and serves as a safeguard against potential issues and deviations, contributing to the reliability of the AI system.

### 4.6. Governance and standardization

There is a shared understanding from AI developers that trustworthy AI should be organizationally driven. Trustworthy AI has been recognized in the literature as the governance framework to document the specific challenges for organizations to address and mitigate on AI systems from a trustworthy and legal lens (Anagnostou et al., 2022; Dignum, 2019). To operationalize trustworthy AI, AI developers need commonality amongst policies, standards, processes and tools to be able to recognize trustworthy issues and take complex trustworthy issues to a deeper expert analysis. We investigated by asking AI developers to highlight the factors that may be barriers to implementing trustworthy AI, which drew the pattern of organizations needing to mandate governance for AI developers to follow:

> Quote #19: "Until it gets to a point where the industry says [everyone must follow] governance and it needs to be part of our regular, daily job. Then you know that's what will make it. Kind of, you know, make it a better situation in the in, in the, in the sense of adopting".

The concepts of governance, standards, framework and policies were referenced a total of 113 times across our research data. There was a clear need by AI developers for a trustworthy AI framework:

> Quote #10: "I guess part of it is just having a framework established. I think that would create your boundaries without having something to align towards from a certification standard, you have to almost have your own program defined within an organization to say these are our risk boundaries. These are thresholds for quality, for performance, our levels of bias, and testing. I think the barriers would be kind of more just having structure in defined in your organization".

The interviews revealed a general consensus regarding the necessity of a reliable framework, standards and guidelines for trustworthy AI. However, it also indicated that organizational pressures could affect the rate at which AI developers adopt a trustworthy AI framework, standards, and guidance:

> Quote #8: "The speed in which want to deliver the product, the business pressure, competition, we kind of bypass some of the policies. To go to the market".

In order to support trustworthy AI in practice, trustworthy AI needs to fit in the organization structure and its subsystems to be a part of the organization's environment (Abedin, 2022).

### 4.7. Further standardization and clear requirements

Our research revealed a gap in the understanding of regulatory considerations among AI developers engage throughout all stages of the AI system lifecycle. AI developers are often driven by technological innovation and the pursuit of efficient solutions, demonstrated a lack of awareness regarding the intricate regulations and compliance requirements that affect their AI systems. AI developers who are unaware of the pertinent regulations may inadvertently design AI systems that fall short of legal and ethical compliance. This can expose organizations to legal liabilities and risks. Therefore, there is a clear need for educational initiatives, training and resources aimed at bridging the awareness gap. These initiatives should help AI developers gain a comprehensive understanding of the regulatory landscape, enabling them to proactively consider and incorporate compliance measures into their AI systems through the AI system lifecycle:

Quote #12: "These are the existing regulations that need to be proved. I think it changes from company to company and case to case. Right maybe before the AI system building process starts if the data scientists could be educated on these regulations that would be good. they will keep in mind all the regulations prior to even build starting the development work".

An additional pattern that emerged from our research was the importance of standardization of trustworthy AI systems' across the entire lifecycle:

- Well-defined trustworthy AI requirements: AI developers expressed a strong need for well-defined trustworthy AI requirements that would serve as guidance during the development process of the AI system. The absence of such clear guidelines can lead to ambiguity and inconsistency in ethical decision-making, potentially resulting in AI systems that infringe of ethical norms (Morley et al., 2020; Raab, 2020).
- Clarity of the types of controls: Our research uncovered a desire for explicit clarity on the types of controls that should be in place to ensure that AI systems operate in a trustworthy manner. Controls in this context are related to accountability, fairness, transparency, explainability and interpretability. Without a clear understanding of the controls, AI developers may struggle to implement ethical safeguards effectively.
- Interest in governance: Our findings indicated a growing interest in governance mechanisms for AI systems. Our research participants recognized the need for oversight and accountability in the AI system lifecycle. The emerging field of governance on AI system seeks to establish frameworks and mechanisms to ensure trustworthy AI systems. As Morley et al. (2020) argue increased awareness of the impact of AI systems has highlighted the inadequacy of traditional 'hard' governance methods, such as regulation and legislation, in protecting individuals, groups, society and the environment. The limitations in 'hard' governance methods, is what led to organizations and governments moving towards 'soft' governance methods, such as ethics strategies, policies, guidelines and frameworks to support enhancing the regulatory landscape of AI (Morley et al., 2020).
- Limited controls in trustworthy AI: In trustworthy AI where controls were limited or non-existent (Morley et al., 2020). This highlights the urgency for more extensive research efforts. Trustworthy AI should not rely on voluntary compliance. Our results displayed the need to have enforceable standards and regulations to ensure that AI systems uphold ethical principles. The need for standardization extends beyond governance and ethical requirements. Our research data also underscored the importance of standardizing AI tool sets, to facilitate the various stages of the AI system lifecycle, including data capture, AI system training, testing and evaluation. A curated list of trustworthy AI tools can empower AI developers to make informed choices that align with the objectives of trustworthy AI. This dual emphasis on standardization, both in terms of ethical principles and AI tools, contributes to trustworthy AI systems.

## 5. Discussion

Several key findings have emerged in this research leading to important recommendations in the field of trustworthy AI. The data we collected from AI developers revealed patterns between the implementation practices used by AI developers and trustworthy AI. Our research has shown that AI developers recognize trustworthy AI is in an immature state with an irregular global regulatory landscape. Furthermore, our findings also highlight the lack of common definitions for trustworthy AI concepts such as interpretability, XAI, and transparency used by AI developers. As a result, AI developers identified that there are limited standards, techniques and tools available to support the implementation of trustworthy AI in practice.

It is evident from our research that AI developers argue trustworthiness in AI systems should be motivated by individuals and organizational efforts. Our research found AI developers highlight the importance of organizations having the responsibility to provide greater standardization of trustworthy AI concepts and practices that are aligned to the global laws and regulations. AI developers' also emphasize the need for clear requirements from the organization on the process of ensuring and maintaining trustworthy AI. Our first recommendation is the need for greater standardization of trustworthy AI concepts and methods in organizations, aligned with research best practices. Organizations need to play a more active role in building trustworthy AI across their organization. We recommend standardizing the trustworthy AI requirements and a means to facilitate consistency in implementing trust in AI systems.

Our findings are consistent with existing literature that revealed trustworthy AI decision-making processes or techniques were often left for AI developers (Abedin, 2022). AI developers expressed a desire for a reliable AI framework, while they emphasized the importance of governance over AI systems. Our second recommendation is investment in techniques and tools to support trustworthy AI across an organization. Whereby, organizations need to invest in building the foundational components of trustworthy AI. This may include governance, training, framework, standards, policies, advisory and guidance from subject matter experts. Despite the usability limitations of trustworthy AI tools (Morley et al., 2020), providing AI developers with the visibility of these trustworthy AI tools and their limitations can help provide further consistency and alignment for AI developers.

In parallel to a finding by Ibanez and Olmeda (2022)'s study, our research also found the importance of ensuring AI developers have trustworthy AI training materials easily available. The training materials must incorporate trustworthy AI throughout all phases of the AI system lifecycle. Our third recommendation is to create training for all roles and responsibilities tied to the AI system lifecycle, this includes internal employees and external customers. This involves providing consistent training to all AI developers in the organization on the approach to trustworthy AI throughout the AI system lifecycle. We encourage comprehensive training to help individuals understand the specific data and AI system requirements relevant to the problems AI developers and other stakeholders aim to address.

The discrepancies in our findings on AI system monitoring during the development and production phases of the AI system lifecycle is a notable area for improvement. To advance the AI system's reliability and robustness, our fourth recommendation is a focus on implementation of systematic and sustained monitoring in AI systems. Implementing monitoring of the AI system's outputs can help AI developers to identify drift in performance or areas of improvement in the AI system. Standardizing monitoring as a requirement would allow for AI developers to proactively assess AI system performance and take timely corrective actions in response to performance degradation or drift.

When it comes to AI systems, data plays a crucial role. Our research has identified several key challenges, such as comprehending the inputs and outputs, type, quantity, sources, and quality, that should be considered in the early phases of AI system development. AI developers in our research underscored the importance of considering trustworthy AI challenges they may face at any point during the problem and development phases of the AI system lifecycle. Our results also highlight the importance of AI developers' needing to have a deep understanding of data when integrating the data into the AI systems. Our fifth recommendation is to build proper data infrastructure in the AI system development phase to support AI developers. In turn, this will help to highlight the challenges associated with managing data restricted by regulatory concerns. Organizations can overcome these challenges, by building strong data infrastructures to support AI developers in creating AI systems and provides the establishment of robust

channels to understand customer needs during problem identification, data collection and analysis phases.

The findings in this paper should act as a guiding framework on how to consider standardizing trustworthy AI in an organizational setting for greater alignment in mitigating the challenges face by AI developers.

## 6. Conclusion

In order to uphold trust in AI systems, it is crucial for AI experts to have access to the proper tools, techniques, and training to implement trustworthy AI research effectively. However, the inconsistent concepts, tools, and practices surrounding trustworthy AI, coupled with a regulatory environment that is rapidly evolving, present significant obstacles to implementation. It is imperative for organizations to proactively invest in building trustworthy AI infrastructures to support implementation and mitigate the risks associated with untrustworthy AI systems before regulations are put in place. In this regard, the responsibilities of trustworthy AI fall on both organizations and individuals, with AI experts playing an important role in the lifecycle stages of AI systems. To address the gap between trustworthy AI research and its practical implementation, it is essential to conduct exploratory research on the perspectives of AI experts . System developer's perspective to trustworthy AI development and deployment can highlight common misconceptions and concepts, test usability and accessibility from a end-user perspective, encourage human-centered design, challenge assumptions and bridge the research gaps in trustworthy AI. The findings of this study will help identify the specific needs and challenges they face in successfully implementing trustworthy AI. With this knowledge, future research can focus on providing implementation tooling and specific trustworthy AI concepts for AI experts.

## CRediT authorship contribution statement

**Carter Cousineau:** Writing – original draft. **Rozita Dara:** Supervision, Writing – review & editing. **Ataharul Chowdhury:** Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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